

AP Physics 101: Kinematics and Dynamics

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Part 1: Kinematics

Workshop Overview Part 1

By the end of the workshop, you will be able to:

- 
- Compare Vector and Scalar Quantities
 - Connect Position Displacement Velocity Acceleration and Time
 - Use Kinematic Equations
 - Interpret Motion Graphs
 - Tackle The Concept of Free Fall Motion

Vector vs Scalar

What is a Vector Quantity?

- Vector quantity measures both magnitude and direction
- For example: Scalar: 3m, 5m/s 10m/s² but Vector: 15m West
5m/s East and so on

Vector quantities: Torque Electric Fields, Acceleration, Force, Weight, Velocity

Scalar Quantities: mass temperature time distance energy

What is Kinematics?

How do we describe motion?

-Kinematics describes an objects:

Position

Displacement

Velocity

Acceleration

Time

Key Variables

x/y = position

Δx = displacement/horizontal displacement

Δy = vertical displacement

t = time

Δt = change in time

v = velocity

v_0/v_i = initial velocity

v/v_f = final velocity

Δv = change in velocity

a = acceleration

g = acceleration of gravity = 9.81 m/s^2



Distance vs Displacement

Example:

- Walk 10m east

- Walk 5m east and 5m west

Distance: 10m 10m

Displacement: 10m 0m

Velocity vs Speed

- Velocity is essentially Speed with direction
- The distinction between scalar and vector units will be very important in interpreting motion graphs.

Acceleration

- Velocity is the rate of change in displacement so how many unit lengths you move per unit time
- Acceleration is the rate of change in velocity, so how many unit velocities you move per unit time.
- So by dimension analysis $v = \text{m/s}$ $a = \text{m/s/s} \rightarrow \text{m/s}^2$
- Note that acceleration can happen without change in speed but cannot happen without change in velocity

Important equations

Kinematics Equations

Equation

Missing Term

$$v = v_0 + at$$

$$x - x_0$$

$$x - x_0 = v_0 t + \frac{1}{2} at^2$$

$$v$$

$$v^2 - v_0^2 = 2a(x - x_0)$$

$$t$$

$$x - x_0 = \frac{1}{2} (v_0 + v) t$$

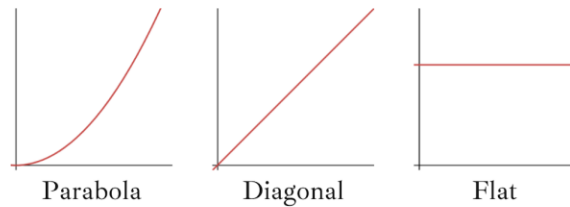
$$a$$

$$x - x_0 = vt - \frac{1}{2} at^2$$

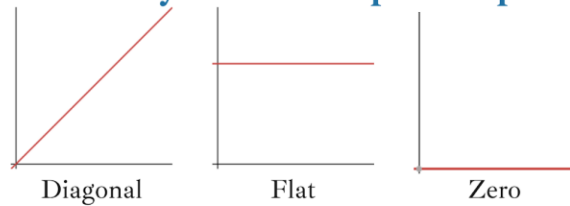
$$v_0$$

Motion Graphs

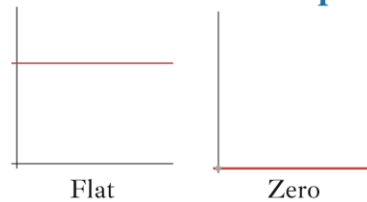
Position-Time Graph Shapes



Velocity-Time Graph Shapes

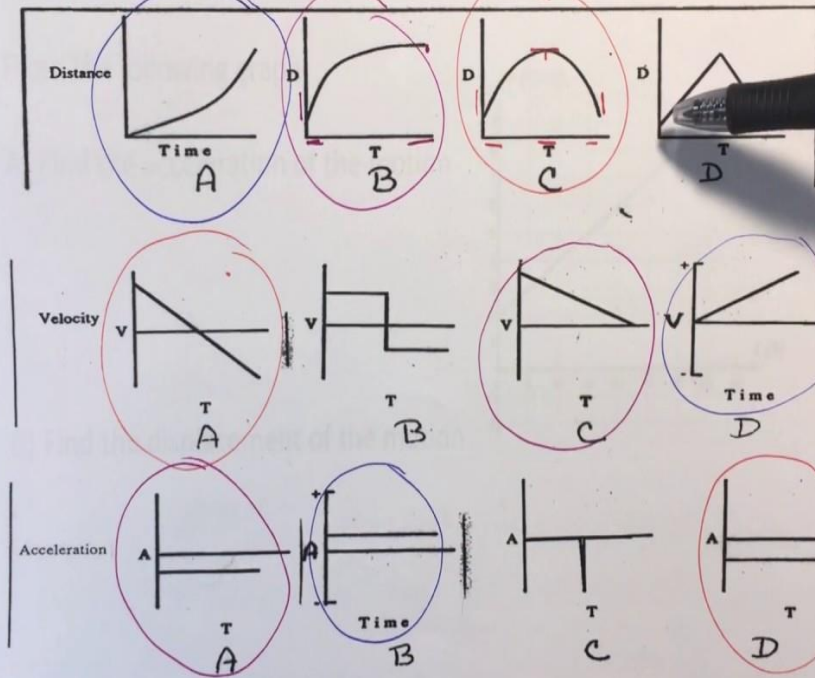


Acceleration-Time Graph Shape



Example #34: Graphs

Match the graphs that have the same motion together



Free-Fall

- In free fall object is thrown into the air and undergoes a constant acceleration equal to g or 9.8.
- Note that motion is upward on the way up but regardless of velocity direction, acceleration vector points downward, thus from the initial launch, object undergoes constant downward a equal to g

Part 2: Dynamics



Workshop Overview Part 2

By the end of the workshop, you will be able to:

- 
- Understand the nature of change of motion
 - Analyze various types of forces
 - Conceptualize the 3 laws of Newton
 - Draw FBDs

What Causes Motion?

- Imagine a chair with wheels....
- To move it you need to push it there isn't a case where a stationary chair suddenly starts moving without external involvement.
- Essentially all objects have the inherent tendency to maintain their state of motion.
- You need to apply force to stop a flying soccer ball and also to kick it across a field.

Newton's First Law

- Inertia: tendency of objects to resist changes in their speed or direction
- An object at rest stays at rest unless acted upon by an **external unbalanced force**

Newton's Second Law

- Force, Mass and Acceleration on a system is related by the equation $\text{Force} = \text{mass} \times \text{acceleration}$
- More Force = more acceleration
- More massive objects = harder to accelerate

Free Body Diagrams 101

- Draw the object only and nothing else
- Draw EVERY FORCE acting on the object
- Label all forces
- Choose positive direction
- Apply Newton's 2nd law
- (optional) some questions may indicate to draw all forces either at the center of mass or the point on the object that the force acts upon

Common Forces in Physics

- Weight (do not confuse with mass)
- Normal Force
- Tension
- Friction
- Applied
- Centripetal
- Gravitational
- Electrical
- Magnetic



Inclined Planes

- Break forces down using components and trigonometric ratios

Newton's Third Law

- Every action results in an equal and opposite reaction force which are called action-reaction pairs
- Difference in mass is what causes acceleration
- Interaction must always occur in a pair

Common AP Mistakes

- Distance $\neq$ displacement
- Speed $\neq$ Velocity
- Gravity always acts downwards even while moving up
- Newton's 3rd law pair cannot form in a single object

Part 3: Review Kahoot!



Part 4: Q&A Time!



Thank you for joining us today!

